

Broader Impact Statement

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My approach to my academic career is fundamentally shaped by my personal and professional experiences, and reinforced through my academic and outreach activities. I believe in creating an environment where students and researchers from all backgrounds feel valued, supported, and empowered to succeed. Through my teaching, research, and service, I strive to reduce barriers, build community, and promote equitable opportunities in academia and beyond.

Personal Background and Academic Service. As a female researcher from a developing country and the first woman in my family to pursue engineering and earn a graduate degree in the United States, I have navigated significant academic and cultural challenges. These experiences give me a unique understanding of the barriers that can hinder student success and fuel my commitment to building supportive, inclusive communities where all students can thrive. This experience has given me a firsthand and unique understanding of the academic and cultural barriers that can hinder student success, and it has motivated me to build the kind of supportive environment that enables all students to thrive.

At the college level, I chair the College of Engineering (CoE) Dean's Graduate Advisory Group, where I translate student feedback into high-impact programs and timely solutions. A key achievement was organizing the inaugural CoE Graduate Research Summit, which featured 50 posters and 10 lightning talks, creating a critical platform for graduate research visibility. I recruited and coordinated 25 faculty reviewers to ensure a rigorous selection process. Building on this success, I launched the CoE Grad Life Series, offering workshops and panels on research ethics, job market preparation, and work-life balance. I also led the efforts to allot dedicated teaching assistant (TA) rooms. Through these efforts, I both advocate for student interests and actively build resources and community for my peers.

In my department, I co-organize the ECE Women Dinner Series, supported by a competitive campus climate grant, and was the first student inductee into the ECE DEI Committee. I helped launch a Tea Time series for informal peer discussions, co-developed a peer advising program pairing freshmen with upper-level mentors, and founded the ECE graduate gender diversity group. Growing from three to over thirty members, the group now holds bi-weekly meetings and retreats, funded through grants I secured, to provide mentorship, community, and professional development for female and non-binary graduate students. I have also advocated for students in crisis, including working with the Dean's office to resolve a conflict over shared lab spaces. My efforts resulted in a new college-wide policy that guarantees equitable access to shared labs. In recognition of these contributions, I received the 2024 UMass CoE DEI Award.

Beyond my institution, I am committed to outreach and mentorship that expands access to STEM. I designed and led a three-week embedded systems course for Springfield High School, introducing seniors from many underrepresented communities to programming and hardware integration. I have mentored students in the UMass Turing Summer Program through lectures and research demos, and I run sessions with elementary school and high school robotics clubs to spark curiosity about emerging technologies. These initiatives aim to make computing accessible to students who might not otherwise have the opportunity to encounter it. I am dedicated to outreach and mentorship that inspires the next generation of leaders.

Research and Broader Impact. *In my research,* I focus on designing safer, more adaptive Mixed Reality (MR) systems that provide secure, fair, and personalized experiences. My work addresses pressing issues such as bias, safety risks, and equitable system design. *In the broader research community,* I have advanced DEI through national roles. As a CRA-E Grad Fellow and editor of the CRA Undergraduate Research

Highlights, I spotlighted the work of students from diverse backgrounds, sharing their unique journeys and contributions. I have mentored underrepresented undergraduates through the NSF REU Computing for an Equitable Energy Transition (CEET) program, guiding them to tackle technical problems with real-world societal impact.

Future Plans

Leadership and Service for Equity and Inclusion. I understand the importance of building initiatives that create lasting institutional change. I will actively engage with my department's existing initiatives, contributing to programs that support students' well-being and promote an inclusive academic culture. Additionally, I will work to strengthen mentoring programs, build bridges between academia and industry, and work with my department to create inclusive networking opportunities for students and faculty to support collaboration and professional growth. Outside academia, I will continue engaging with K-12 outreach and professional organizations to inspire underrepresented groups to pursue STEM careers. By extending efforts beyond my lab and classroom, I aim to foster a culture of equity and inclusion at multiple levels of academia and the broader community.

Research for Societal Impact and Inclusivity. The problems I choose to address in my research will reflect my commitment to societal impact and inclusivity. I will focus on challenges relevant to underserved and marginalized communities, such as designing personalized and accessible MR systems, improving sustainability in technology, and addressing biases in computing systems. These research directions align with my belief that technology should benefit everyone equitably rather than perpetuate existing inequalities. Furthermore, I will adopt open science practices, sharing datasets, tools, and methodologies to democratize research participation and amplify my work's impact.